LAB-04
DEWAGEDARA D.M.E.S.
E/21/087

11) Write a nested query to list the details of the movies directed by a director

A. who is also a reviewer.

```
mysql> SELECT *
    -> FROM
    ->  movie
    -> WHERE
    ->  movie.director IN (SELECT reviewer_name FROM reviewer);
+----------+---------+------+---------------+
| movie_id | title   | year | director      |
+----------+---------+------+---------------+
|      105 | Titanic | 1997 | James Cameron |
|      107 | Avatar  | 2009 | James Cameron |
+----------+---------+------+---------------+
2 rows in set (0.00 sec)
```

B. who is not a reviewer.

```
mysql> SELECT *
    -> FROM
    ->  movie
    -> WHERE
    ->  movie.director NOT IN (SELECT reviewer_name FROM reviewer);
+----------+------------------------+------+------------------+
| movie_id | title                  | year | director         |
+----------+------------------------+------+------------------+
|      101 | Gone with the Wind     | 1939 | Victor Fleming   |
|      102 | Star Wars              | 1977 | George Lucas     |
|      103 | The Sound of Music     | 1965 | Robert Wise      |
|      104 | E.T.                   | 1982 | Steven Spielberg |
|      108 | Raiders of the Lost Ark | 1981 | Steven Spielberg |
+----------+------------------------+------+------------------+
5 rows in set (0.01 sec)
```

12) Write a nested query to list the details of the movie ratings,

A. reviewed by the reviewer ‘Sarah Martinez’.

```
mysql> SELECT *
    -> FROM
    ->  rating
    -> WHERE
    ->  reviewer_id=(SELECT reviewer_id FROM reviewer WHERE reviewer_name='Sarah Martinez');
+-------------+----------+-------+-------------+
| reviewer_id | movie_id | stars | rating_date |
+-------------+----------+-------+-------------+
|         201 |      101 |     2 | 2011-01-22  |
+-------------+----------+-------+-------------+
1 row in set (0.00 sec)
```

B. not reviewed by the reviewer ‘Sarah Martinez’.

```
mysql>
mysql> SELECT *
    -> FROM
    ->  rating
    -> WHERE
    ->  reviewer_id NOT IN (SELECT reviewer_id FROM reviewer WHERE reviewer_name='Sarah Martinez');
+-------------+----------+-------+-------------+
| reviewer_id | movie_id | stars | rating_date |
+-------------+----------+-------+-------------+
|         202 |      106 |     4 | NULL        |
|         203 |      103 |     2 | 2011-01-20  |
|         203 |      108 |     4 | 2011-01-12  |
|         204 |      101 |     3 | 2011-01-09  |
|         205 |      103 |     3 | 2011-01-27  |
|         205 |      104 |     2 | 2011-01-22  |
|         205 |      108 |     4 | NULL        |
|         206 |      106 |     5 | 2011-01-19  |
|         206 |      107 |     3 | 2011-01-15  |
|         207 |      107 |     5 | 2011-01-20  |
|         208 |      104 |     3 | 2011-01-02  |
+-------------+----------+-------+-------------+
11 rows in set (0.00 sec)
```

13) Find all the years that have a movie that received a rating of 4 or 5 and sort them in increasing order of the year. Write,

A.non-nested query.

```
mysql>
mysql> SELECT
    ->   DISTINCT movie.year
    -> FROM
    ->  movie
    -> JOIN
    ->  rating ON movie.movie_id=rating.movie_id
    -> WHERE
    ->  rating.stars = 4 OR rating.stars = 5
    -> ORDER BY
    ->  movie.year ASC;
+------+
| year |
+------+
| 1937 |
| 1981 |
| 2009 |
+------+
3 rows in set (0.00 sec)
```

B.a nested query

```
mysql> SELECT
    ->   DISTINCT movie.year
    -> FROM
    ->  movie
    -> WHERE
    ->  movie_id IN (SELECT movie_id FROM rating WHERE stars=4 or stars=5);
+------+
| year |
+------+
| 1937 |
| 1981 |
| 2009 |
+------+
3 rows in set (0.00 sec)
```

14) Some reviewers did not provide a date with their rating. Find the names of all reviewers who have a NULL value for the date. Write a nested query

```
mysql> SELECT
    ->  reviewer_name
    -> FROM
    ->  reviewer
    -> WHERE
    ->  reviewer_id IN (SELECT reviewer_id FROM rating WHERE rating_date IS NULL);
+---------------+
| reviewer_name |
+---------------+
| Daniel Lewis  |
| Chris Jackson |
+---------------+
2 rows in set (0.00 sec)
```

15)List the movie titles and average ratings, from the highest-rated to lowest-rated. If two or more movies have the same average rating, list them in alphabetical order.

```
mysql>
mysql> SELECT
    ->  movie.title,
    ->  AVG(rating.stars) AS average_rating
    -> FROM
    ->  movie
    -> JOIN
    ->  rating ON movie.movie_id = rating.movie_id
    -> GROUP BY
    ->  movie.title
    -> ORDER BY
    ->  average_rating DESC,
    ->  movie.title ASC;
+-------------------------+----------------+
| title                   | average_rating |
+-------------------------+----------------+
| Snow White              |         4.5000 |
| Avatar                  |         4.0000 |
| Raiders of the Lost Ark |         4.0000 |
| E.T.                    |         2.5000 |
| Gone with the Wind      |         2.5000 |
| The Sound of Music      |         2.5000 |
+-------------------------+----------------+
6 rows in set (0.00 sec)
```

16) Remove all ratings where the movie's year is before 1970 or after 2000

```
mysql>
mysql> DELETE FROM
    ->  rating
    -> WHERE
    ->  movie_id IN (SELECT movie_id FROM movie WHERE year<1970 OR year > 2000 );
Query OK, 8 rows affected (0.01 sec)

mysql> SELECT * FROM rating;
+-------------+----------+-------+-------------+
| reviewer_id | movie_id | stars | rating_date |
+-------------+----------+-------+-------------+
|         203 |      108 |     4 | 2011-01-12  |
|         205 |      104 |     2 | 2011-01-22  |
|         205 |      108 |     4 | NULL        |
|         208 |      104 |     3 | 2011-01-02  |
+-------------+----------+-------+-------------+
4 rows in set (0.00 sec)
```

17) Remove all ratings where the rating date is NULL

```
mysql>
mysql> DELETE FROM
    ->  rating
    -> WHERE
    ->  rating_date IS NULL;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM rating;
+-------------+----------+-------+-------------+
| reviewer_id | movie_id | stars | rating_date |
+-------------+----------+-------+-------------+
|         203 |      108 |     4 | 2011-01-12  |
|         205 |      104 |     2 | 2011-01-22  |
|         208 |      104 |     3 | 2011-01-02  |
+-------------+----------+-------+-------------+
3 rows in set (0.00 sec)
```